

DATA AND MEASUREMENT

SESSION 3
COMP6140001 – DATA MINING

SUBJECT MATTER EXPERT
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LEARNING OBJECTIVE

L01 : Explain concept of data, data warehouse, data preprocessing, and data mining



OUTLINE

- Data Type
- Statistic of Data

DATA TYPE



DATA

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
 - Attribute is also known as variable, field, characteristic, or feature
 - Examples: eye color of a person, temperature, customerID, name, etc.
- A collection of attributes describe an object
 - Object is also known as record, point, case, sample, entity, or instance

		Attributes			
		<i>Tid</i>	Refund	Marital Status	Taxable Income
Objects	1	Yes	Single	125K	No
	2	No	Married	100K	No
	3	No	Single	70K	No
	4	Yes	Married	120K	No
	5	No	Divorced	95K	Yes
	6	No	Married	60K	No
	7	Yes	Divorced	220K	No
	8	No	Single	85K	Yes
	9	No	Married	75K	No
	10	No	Single	90K	Yes



ATTRIBUTE

Types of attributes:

- Nominal
 - Examples: ID numbers, eye color, zip codes
 - Ordinal
 - Examples: rankings (taste of potato chips on a scale from 1-10), height (tall, medium, short), professional rank (assistant, associate, professor)
- Categorical**
- Numeric: Interval-scaled
 - Examples: calendar dates
 - Numeric: Ratio-scaled
 - Examples: monetary quantities, counts, age, mass, length, electrical current
- Numeric**



ATTRIBUTE (2)

List of attribute value **properties**:

- Distinctness : $= \neq$
- Order : $< >$
- Addition : $+ -$
- Multiplication : $\times \div$

Properties of attribute **types**:

Attribute Type	Value Properties
Nominal	Distinctness
Ordinal	Distinctness & order
Numeric (Interval)	Distinctness, order, & addition
Numeric (Ratio)	Distinctness, order, addition, & multiplication



ATTRIBUTE (3)

Discrete vs. Continuous Attributes:

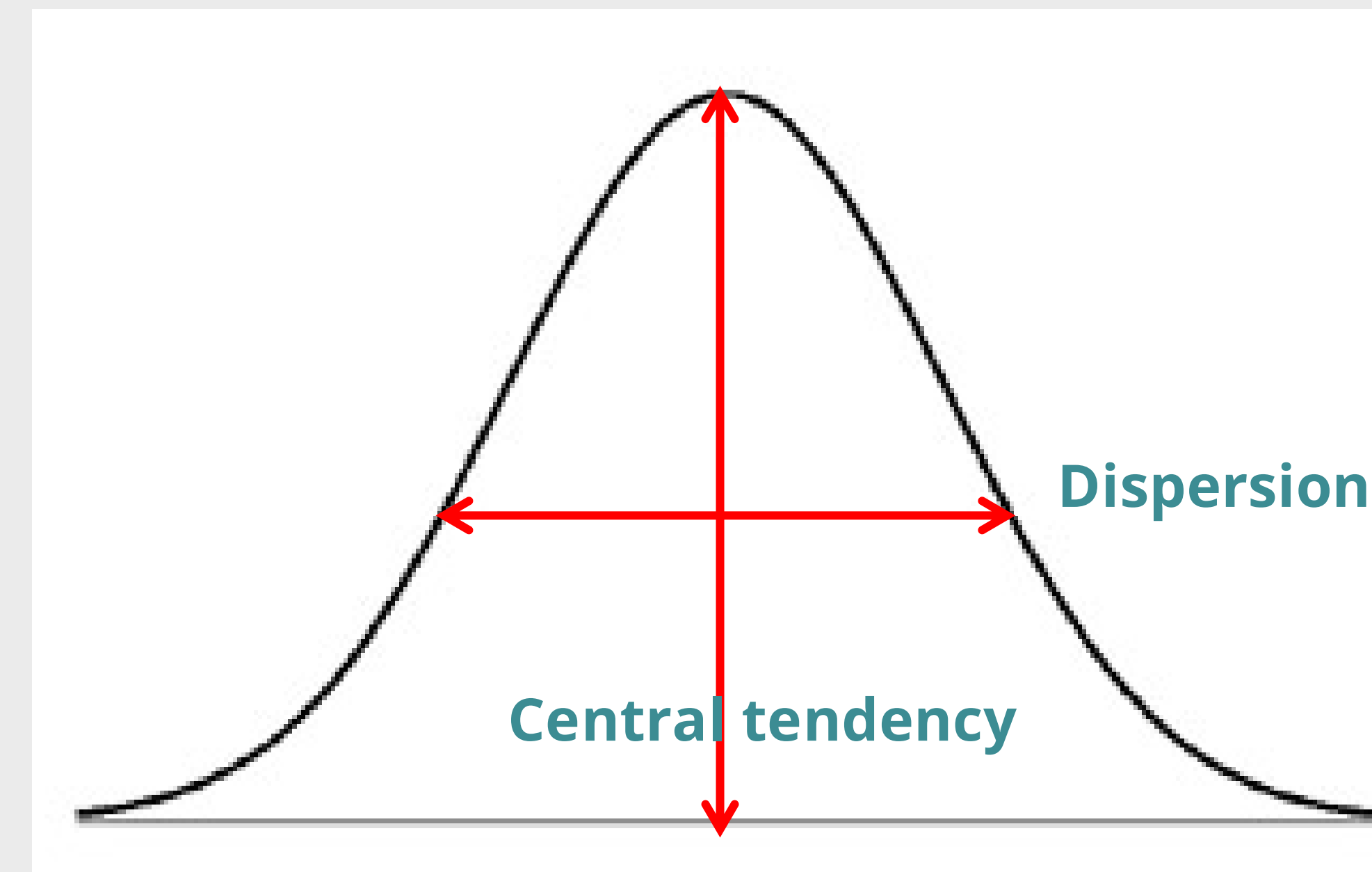
Categories	Discrete Attribute	Continuous Attribute
Attribute values	Finite or countably infinite set of values	Real numbers
Represented as	Integer variables	Floating-point variables
Notes	Binary attributes are a special case	Real values can only be measured and represented using a finite number of digits
Examples	Zip codes, counts, or the set of words in a collection of documents	Temperature, height, or weight

STATISTIC OF DATA



BASIC STATISTICAL DESCRIPTION OF DATA

- Motivation
 - To better understand the data: central tendency, variation, and spread
- Central tendency characteristics
 - Mean, median, and mode
- Data dispersion characteristics
 - Quantiles, interquartile range (IQR), and variance





CENTRAL TENDENCY

Mean (algebraic measure):

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **Weighted** arithmetic mean

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

- Trimmed mean:
chopping extreme values



CENTRAL TENDENCY

Median: middle value if odd number of values, or average of the middle two values otherwise

- Estimated by interpolation (for *grouped data*):

<i>age</i>	<i>frequency</i>
1–5	200
6–15	450
16–20	300
21–50	1500
51–80	700
81–110	44

Approximate
median

Sum before the median interval

Interval width ($L_2 - L_1$)

$$median = L_1 + \left(\frac{\frac{n}{2} - (\sum freq) l}{freq_{median}} \right) width$$

Low interval limit



CENTRAL TENDENCY (2)

- **Mode:** value that occurs the most frequently in the data
 - Unimodal, bimodal, trimodal
 - Empirical formula for unimodal (asymmetrical):

$$mean - mode \approx 3 \times (mean - median)$$

- **Midrange:** average value of the largest and smallest values in the data

$$midrange = \frac{max - min}{2}$$

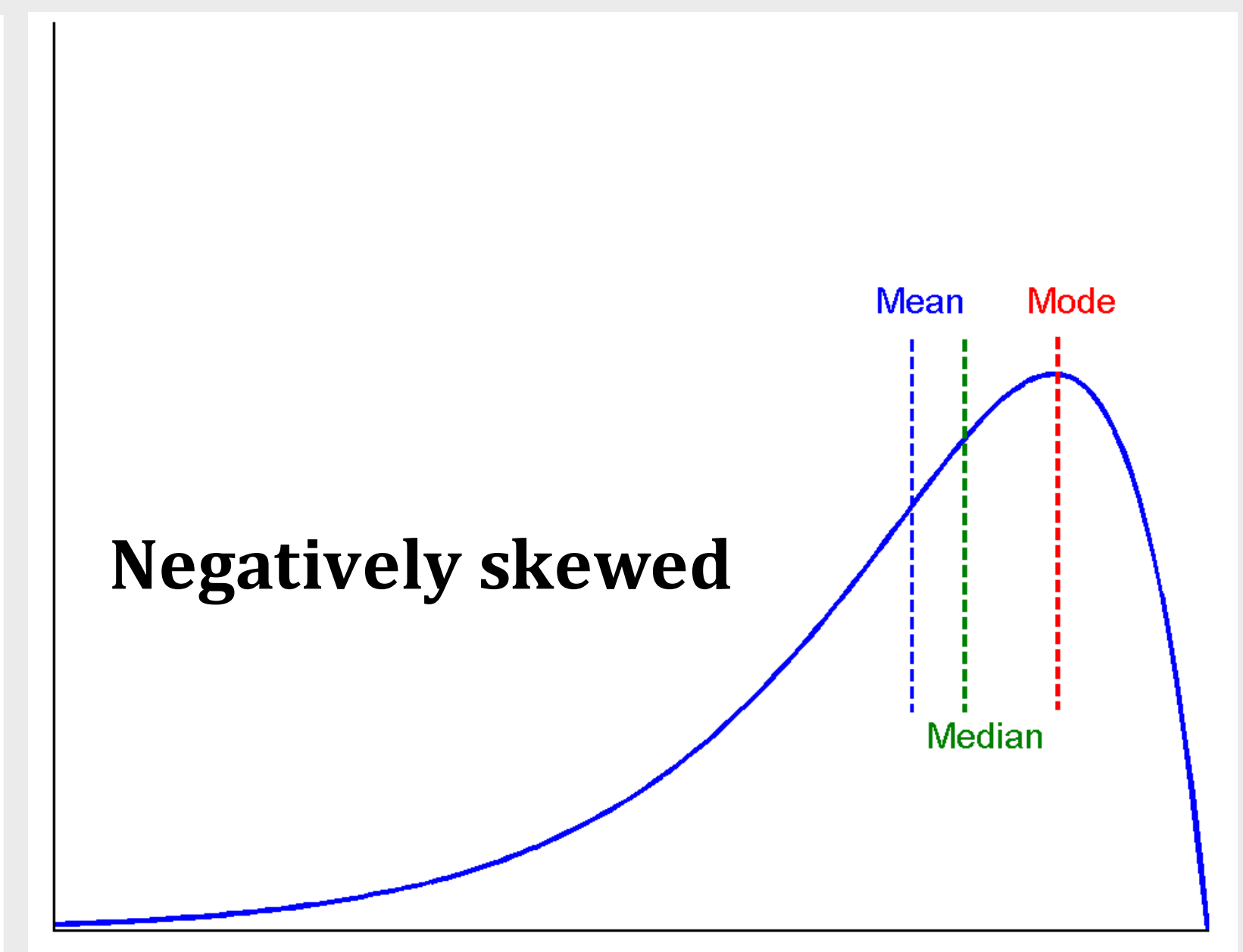
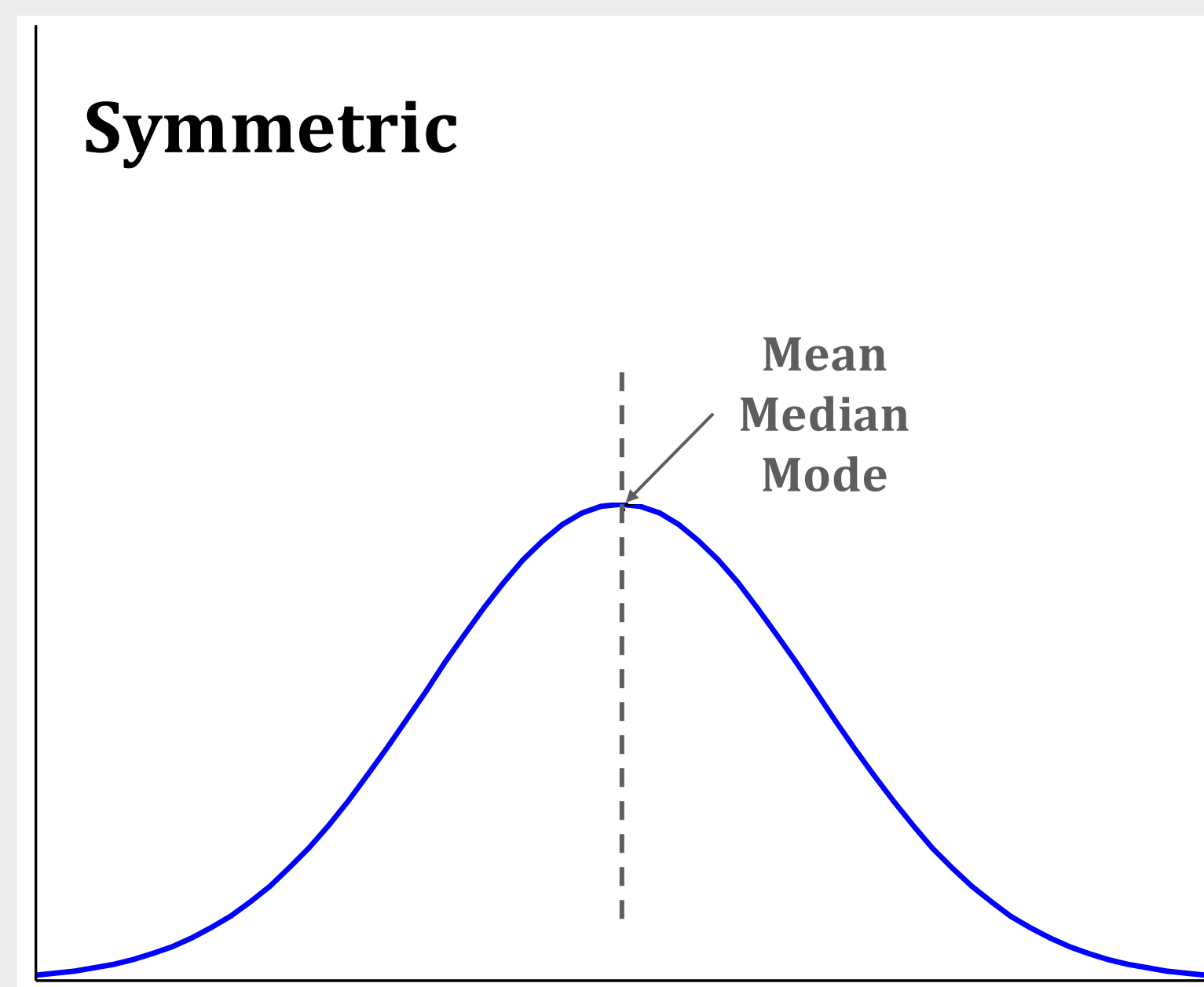
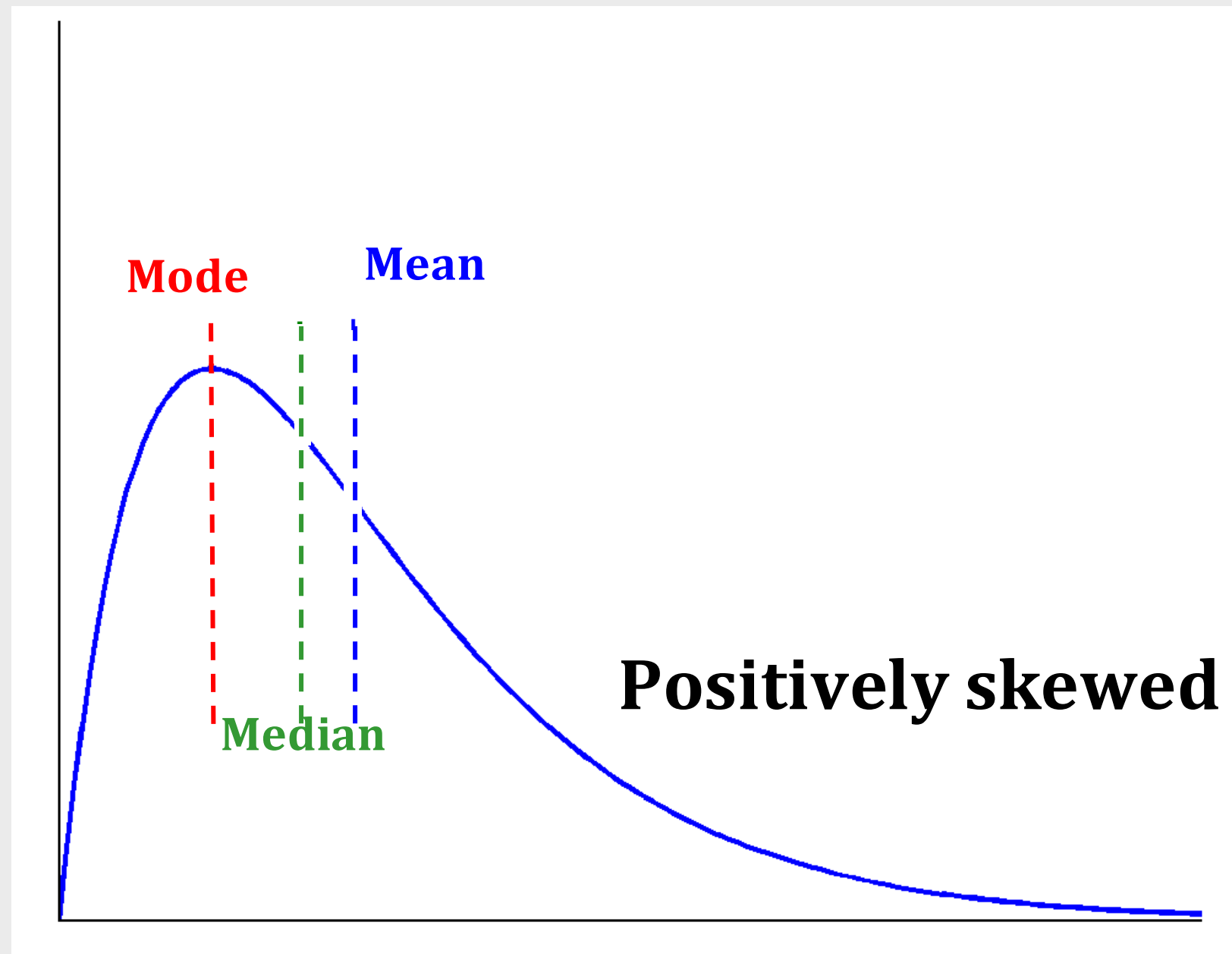
- Example case:

Mean = 532.33
Median = 900
Mode = 1500
Midrange = 728

<i>age</i>	<i>frequency</i>
1–5	200
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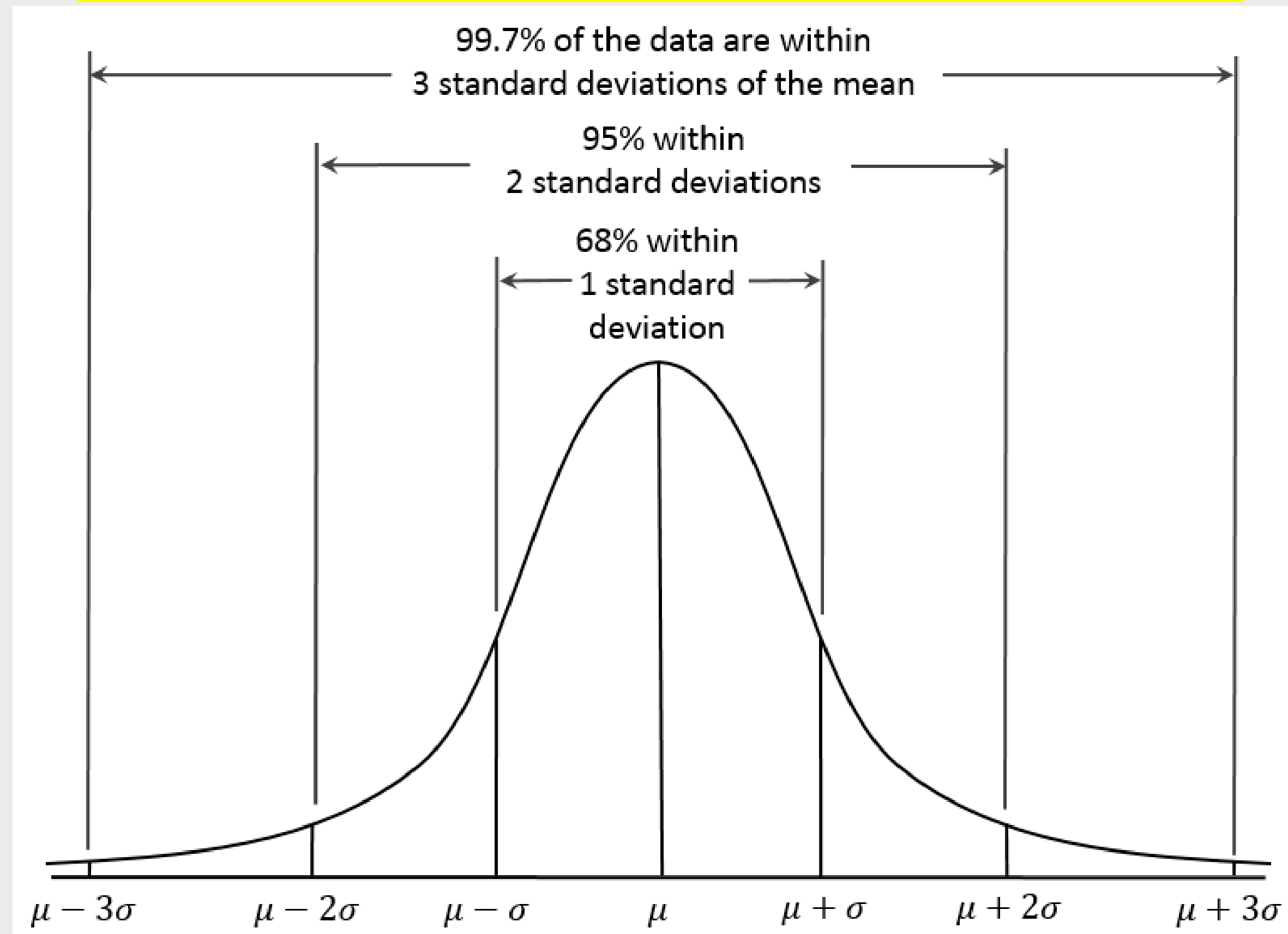
SYMMETRIC VS. SKEWED DATA





PROPERTIES OF NORMAL DISTRIBUTION CURVE

← — — Represent data dispersion, spread — — — →



Represent central
tendency



DISPERSION OF DATA

- **Range:** the difference between the largest (max) and the smallest (min) values.
- **Quartiles:** Q1 (25th percentile), Q3 (75th percentile)
- **Inter-quartile range:** $IQR = Q3 - Q1$
- **Five number summary** of a distribution: min, Q1, median, Q3, max
- **Boxplot:** ends of the box are the quartiles; median is marked; add whiskers, and plot outliers individually
- **Outlier:** usually, a value higher/lower than $1.5 \times IQR$



DISPERSION OF DATA (2)

- **Variance:** algebraic, scalable computation

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^n x_i^2 \right) - \bar{x}^2$$

- **Standard deviation** σ is the square root of variance σ^2

$$\sigma = \sqrt{\sigma^2}$$



CORRELATION ANALYSIS (FOR CATEGORICAL DATA)

- **χ^2 (chi-square) test:**

$$\chi^2 = \sum_i^n \frac{\overset{\text{observed}}{\downarrow} (O_i - E_i)^2}{\underset{\text{expected}}{E_i}}$$

- Null hypothesis: The two distributions are independent
- The cells that contribute the most to the χ^2 value are those whose actual count is very different from the expected count
- The larger the χ^2 value, the more likely the variables are related



CHI-SQUARE CALCULATION: AN EXAMPLE

	Play chess	Not play chess	Sum (row)
Like science fiction	250 (X1)	200 (X2)	450
Not like science fiction	50 (X3)	1000 (X4)	1050
Sum(col.)	300	1200	1500

How to derive 90?
 $450/1500 * 300 = 90$

- χ^2 (chi-square) calculation (numbers in parenthesis are expected counts calculated based on the data distribution in the two categories)

$$\chi^2 = \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} = 507.93$$

We can reject the null hypothesis of independence at a confidence level of 0.001

- It shows that like_science_fiction and play_chess are correlated in the group



CHI-SQUARE CALCULATION: AN EXAMPLE

	A	B	C	D	Sum (row)
1					200
0					1000
Sum(col.)	300	300	300	300	1200

- Degree of freedom
 - $(\text{\#categories_in_variable_A} - 1)(\text{\#categories_in_variable_B} - 1)$
 - number of values that are free to vary

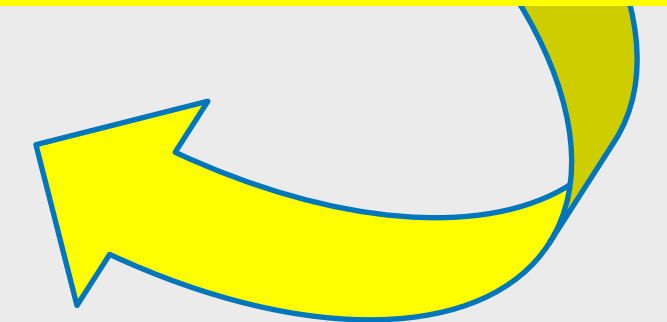


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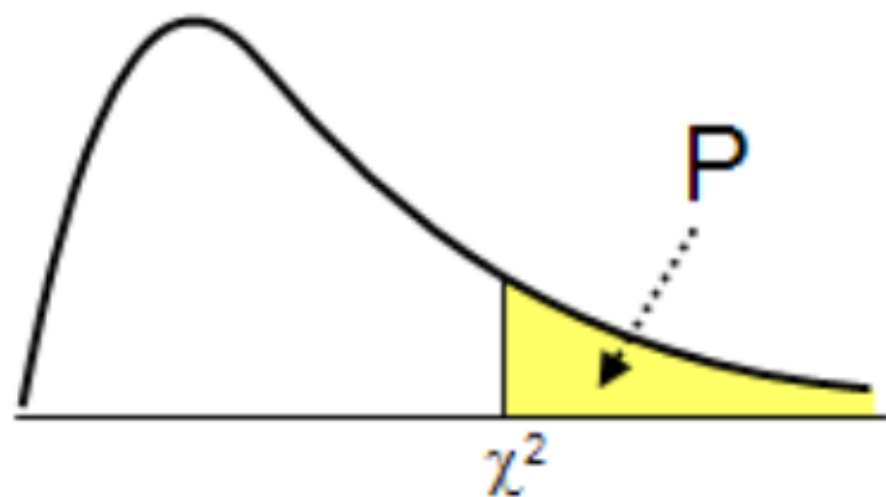
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- Degree of freedom=?

Values of the Chi-squared distribution



	P										
DF	0.995	0.975	0.20	0.10	0.05	0.025	0.02	0.01	0.005	0.002	0.001
1	0.0000393	0.000982	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.550	10.828
2	0.0100	0.0506	3.219	4.605	5.991	7.378	7.824	9.210	10.597	12.429	13.816
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.345	12.838	14.796	16.266
4	0.207	0.484	5.989	7.779	9.488	11.143	11.668	13.277	14.860	16.924	18.467
5	0.412	0.831	7.289	9.236	11.070	12.833	13.388	15.086	16.750	18.907	20.515
6	0.676	1.237	8.558	10.645	12.592	14.449	15.033	16.812	18.548	20.791	22.458



CORRELATION BETWEEN TWO NUMERICAL VARIABLES

- **Correlation** between two variables X_1 and X_2 is the standard covariance, obtained by normalizing the covariance with the standard deviation of each variable.

$$\rho_{12} = \frac{\sigma_{12}}{\sigma_1 \sigma_2} = \frac{\sigma_{12}}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

- **Sample correlation** for two attributes X_1 and X_2 :

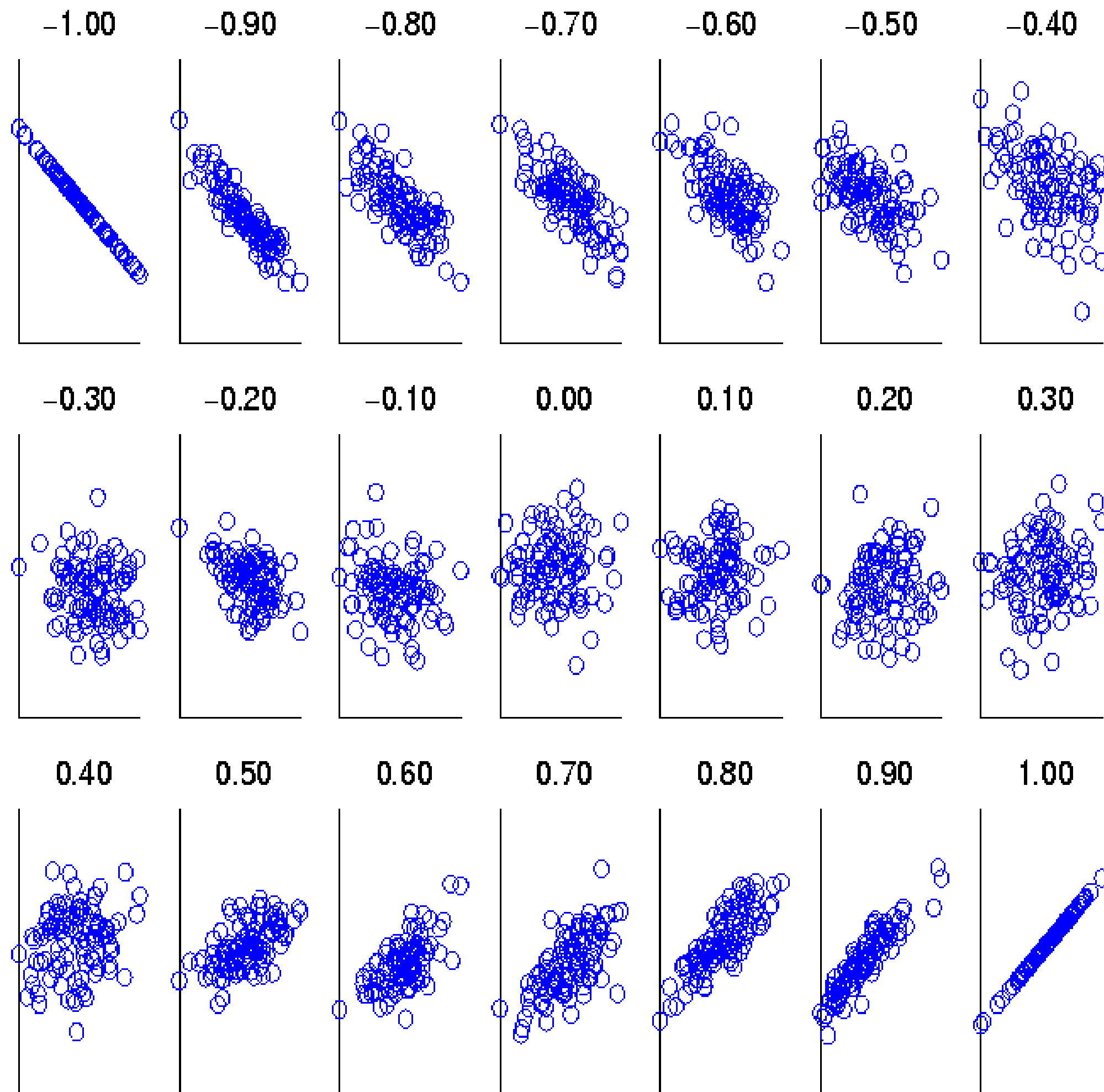
$$\hat{\rho}_{12} = \frac{\hat{\sigma}_{12}}{\hat{\sigma}_1 \hat{\sigma}_2} = \frac{\sum_{i=1}^n (x_{i1} - \hat{\mu}_1) (x_{i2} - \hat{\mu}_2)}{\sqrt{\sum_{i=1}^n (x_{i1} - \hat{\mu}_1)^2 \sum_{i=1}^n (x_{i2} - \hat{\mu}_2)^2}}$$

where n is the number of tuples, μ_1 and μ_2 are the respective means of X_1 and X_2 , σ_1 and σ_2 are the respective standard deviation of X_1 and x_2

- If $\rho_{12} > 0$: A and B are positively correlated (X_1 's values increase as X_2 's)
 - The higher, the stronger correlation
- If $\rho_{12} = 0$: independent (under the same assumption as discussed in co-variance)
- If $\rho_{12} < 0$: negatively correlated



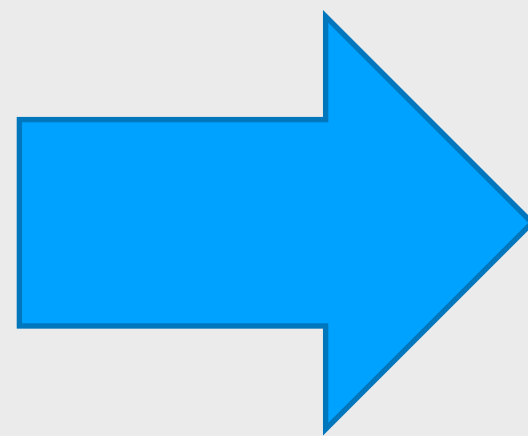
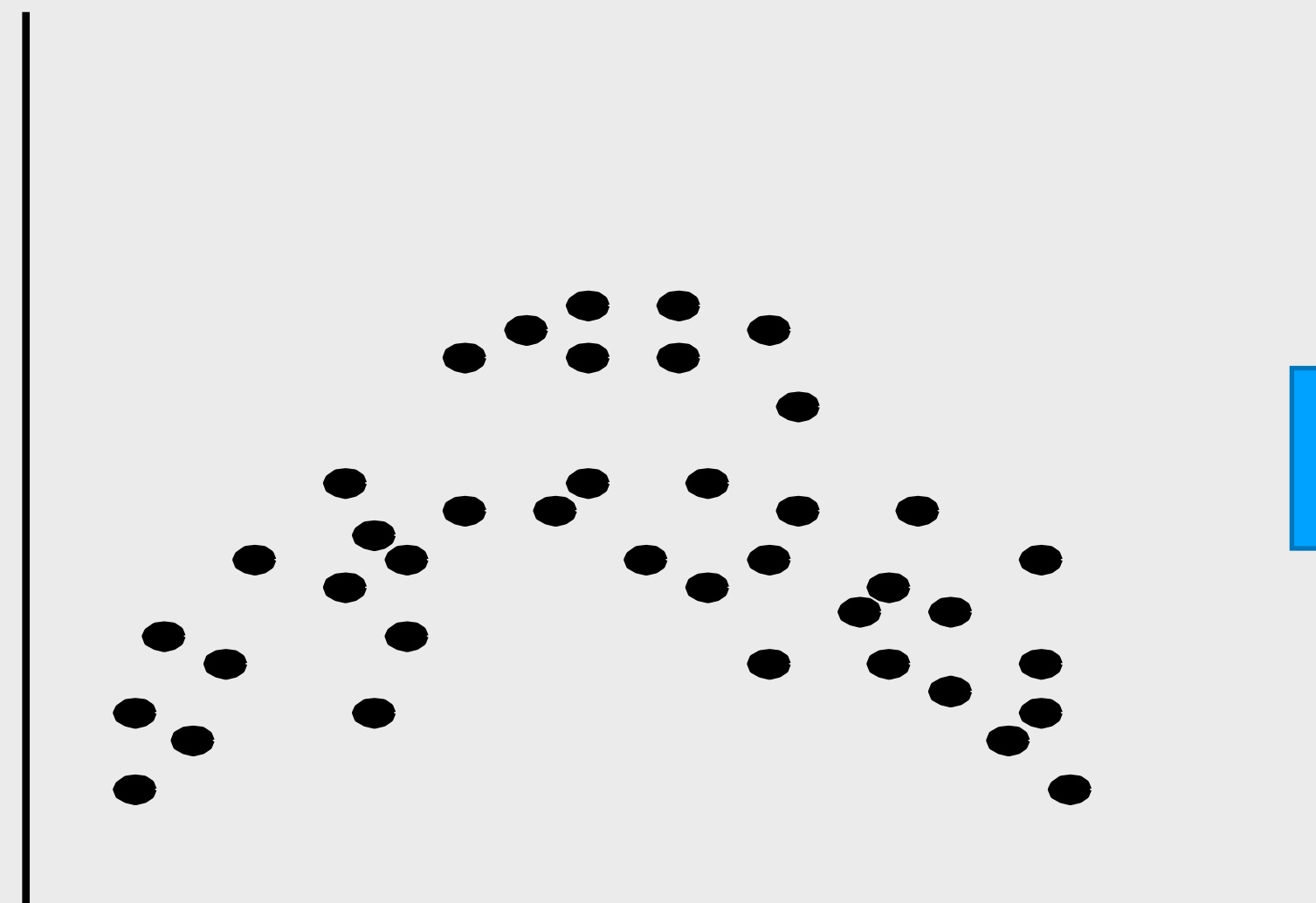
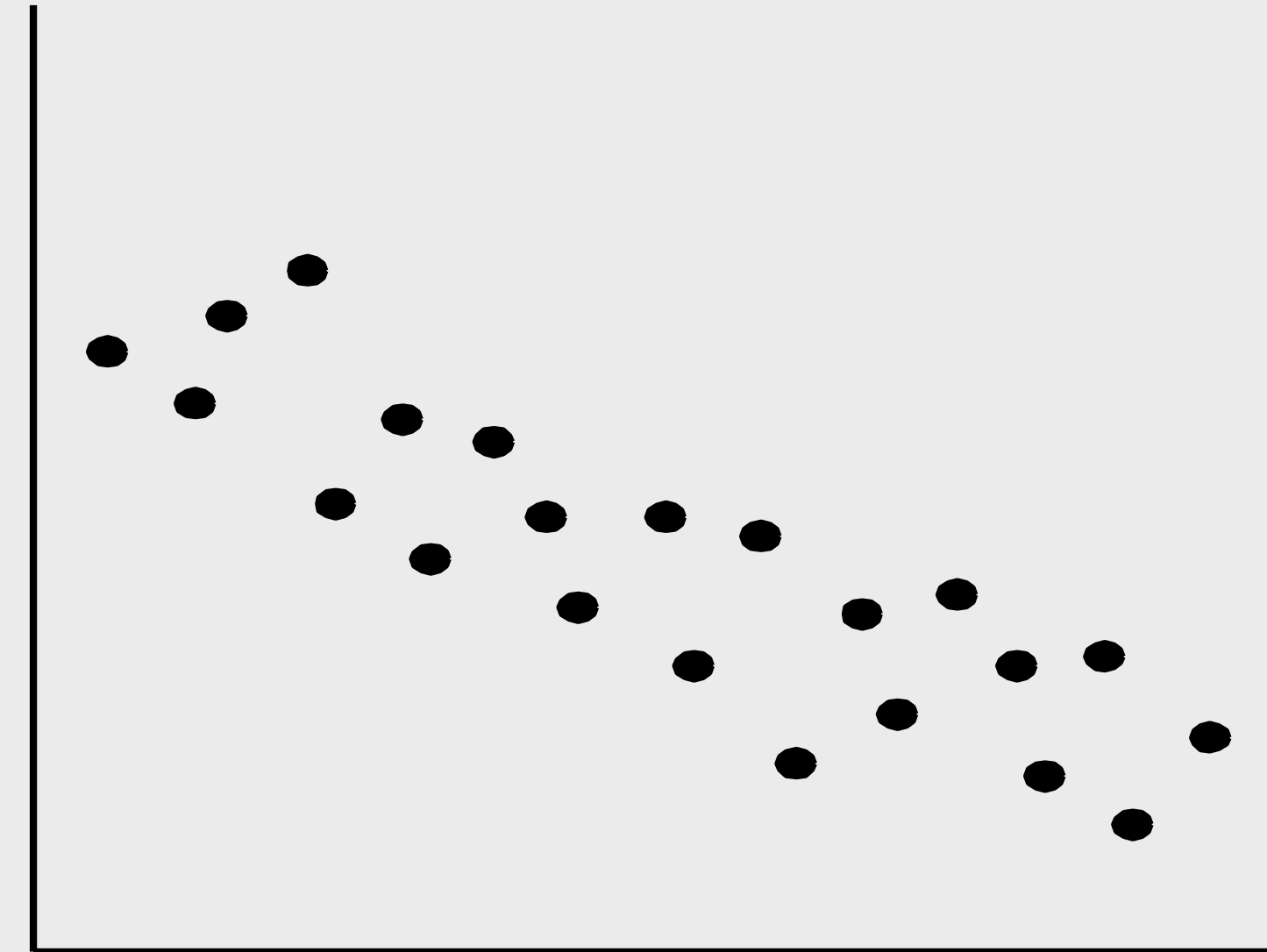
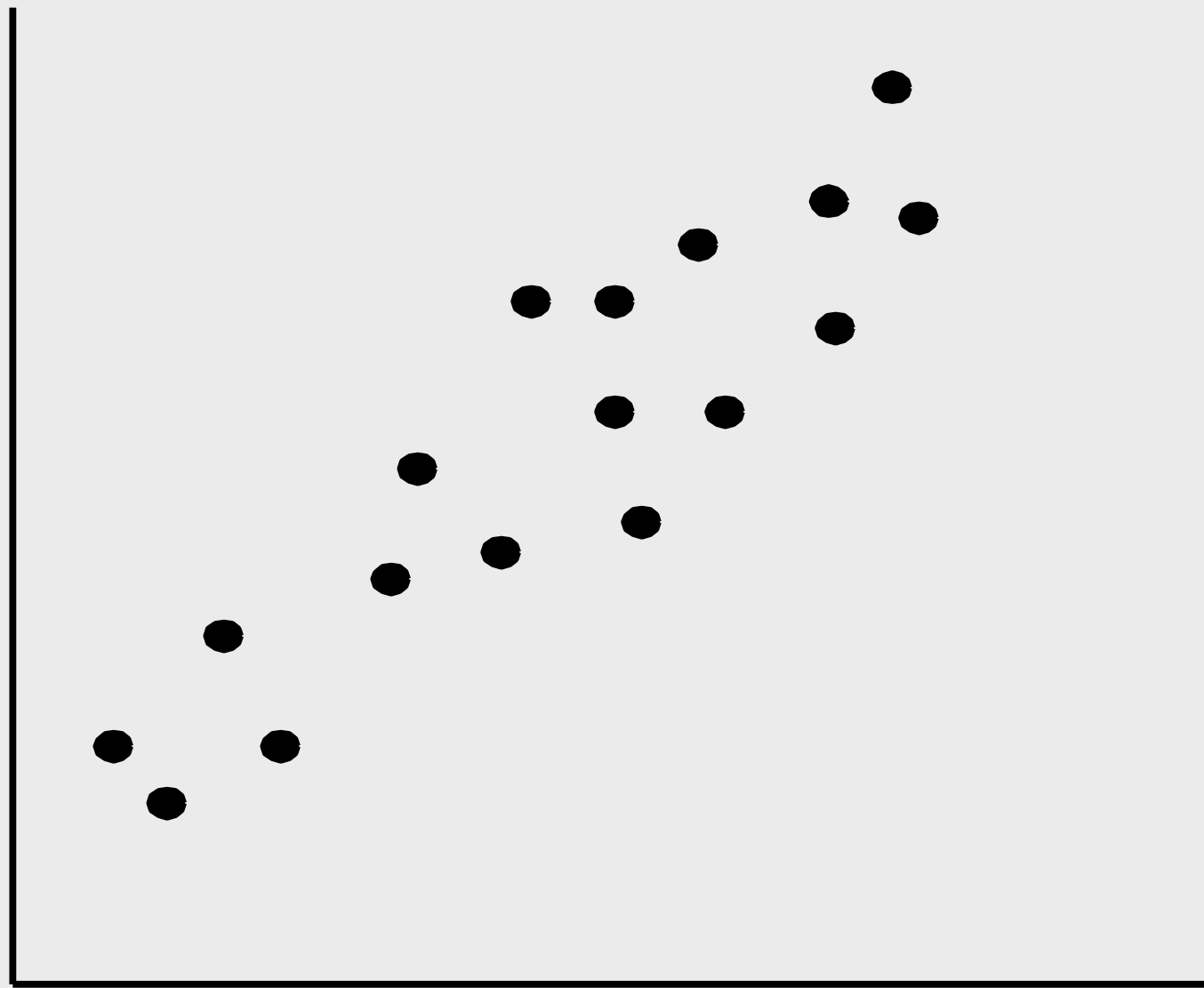
VISUALIZING CHANGES OF CORRELATION COEFFICIENT



- Correlation coefficient value range: $[-1, 1]$
- A set of scatter plots shows sets of points and their correlation coefficients changing from -1 to 1



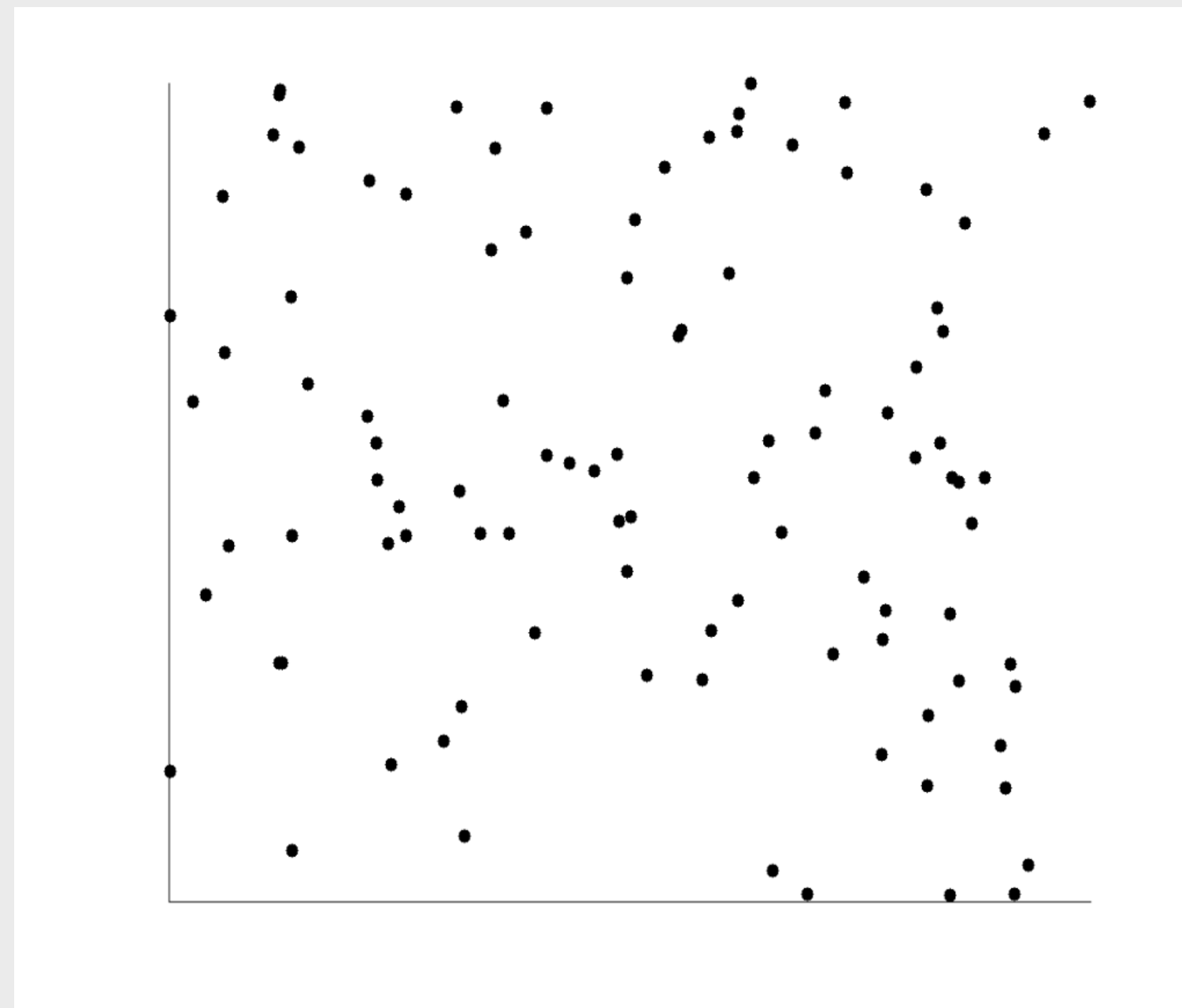
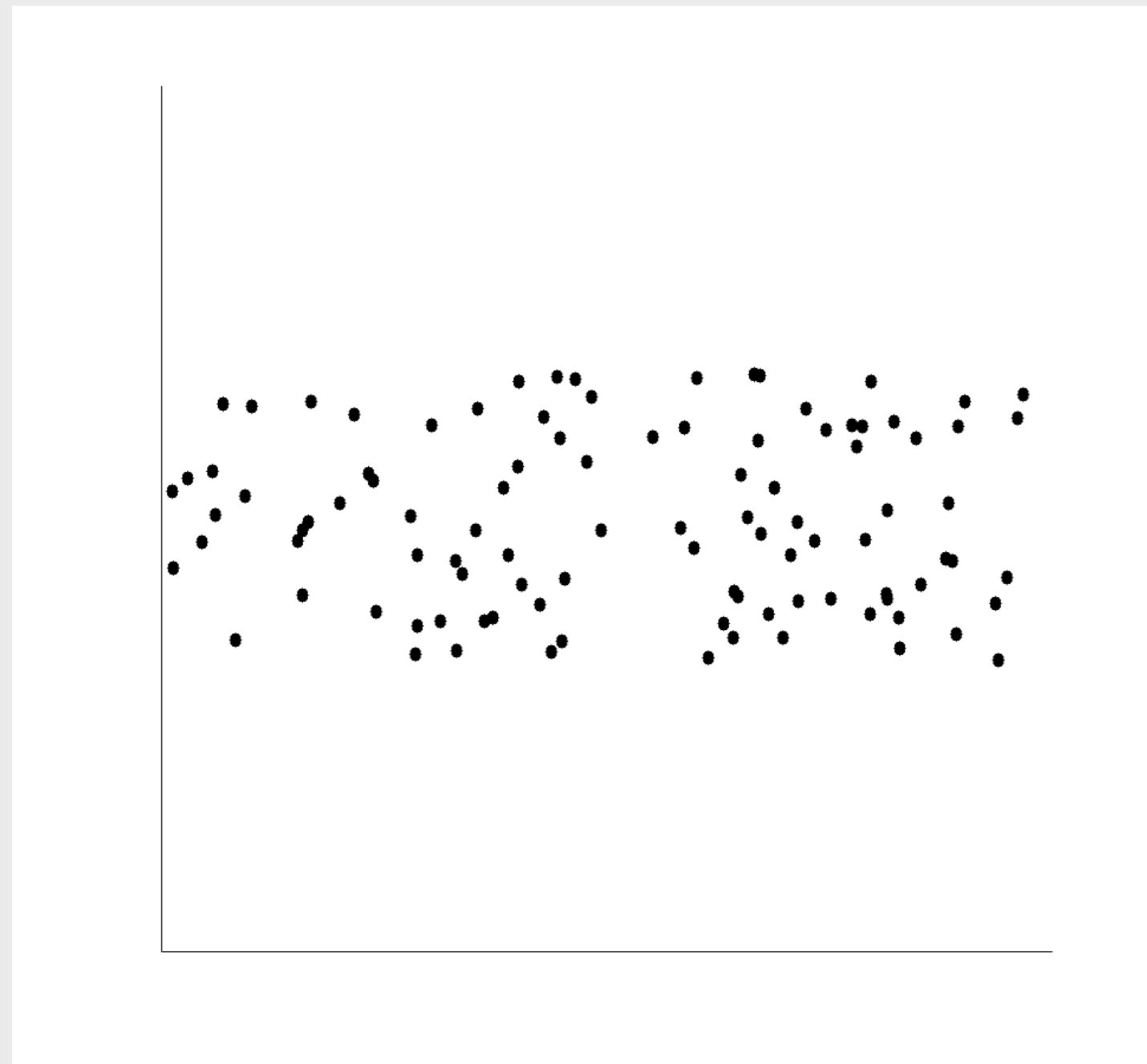
POSITIVELY AND NEGATIVELY CORRELATED DATA



- The left half fragment is positively correlated
- The right half is negative correlated



UNCORRELATED DATA





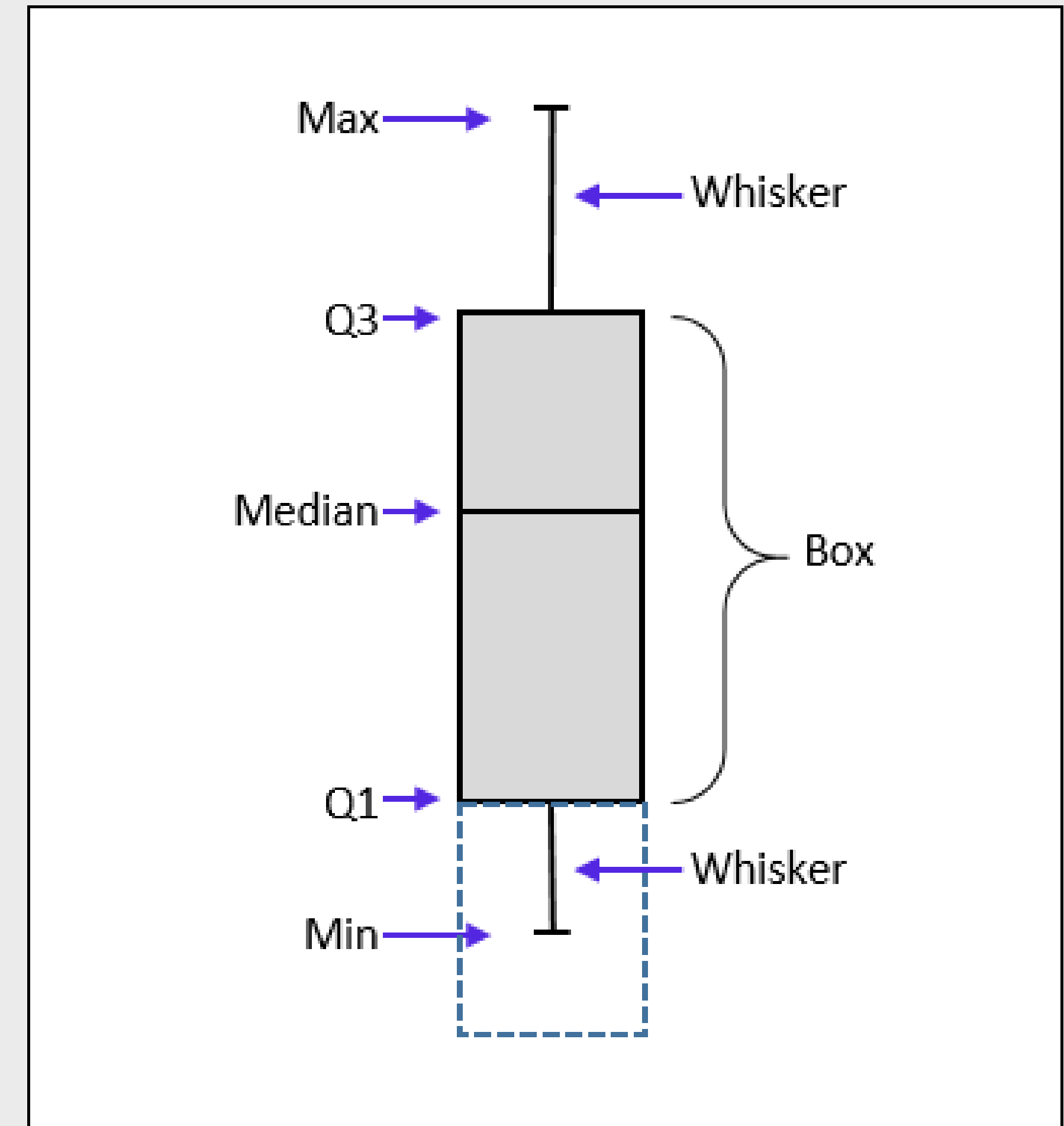
GRAPHIC DISPLAYS OF BASIC STATISTICAL DESCRIPTIONS

- **Boxplot:** graphic display of five-number summary
- **Histogram:** x-axis are values, y-axis represents frequencies (f)
- **Quantile plot:** each value x_i is paired with f_i indicating that approximately 100% % of data are $\leq x_i$
- **Quantile-quantile (q-q) plot:** graphs the quantiles of one univariant distribution against the corresponding quantiles of another
- **Scatter plot:** each pair of values is a pair of coordinates and plotted as points in the plane



BOXPLOT

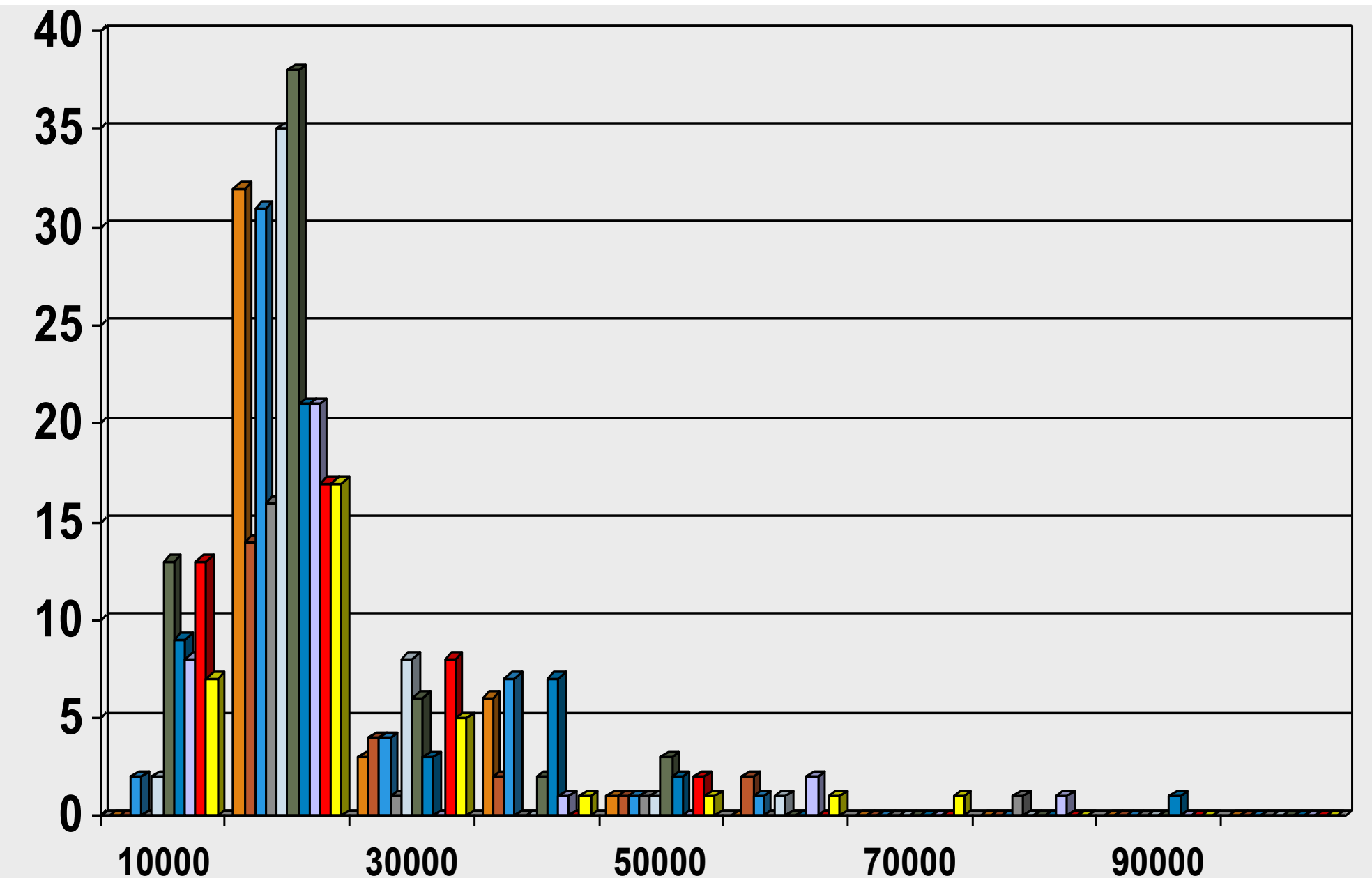
- Quartiles: Q1 (25th percentile), Q3 (75th percentile)
- Inter-quartile range: $IQR = Q3 - Q1$
- Five number summary: min, Q1, median, Q3, max
- Data is represented with a box
 - Q_1 , Q_3 , IQR: The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
 - Median (Q_2) is marked by a line within the box
 - Whiskers: two lines outside the box extended to minimum and maximum
 - Outliers: points beyond a specified outlier threshold, plotted individually. Usually, a value higher/lower than $1.5 \times IQR$



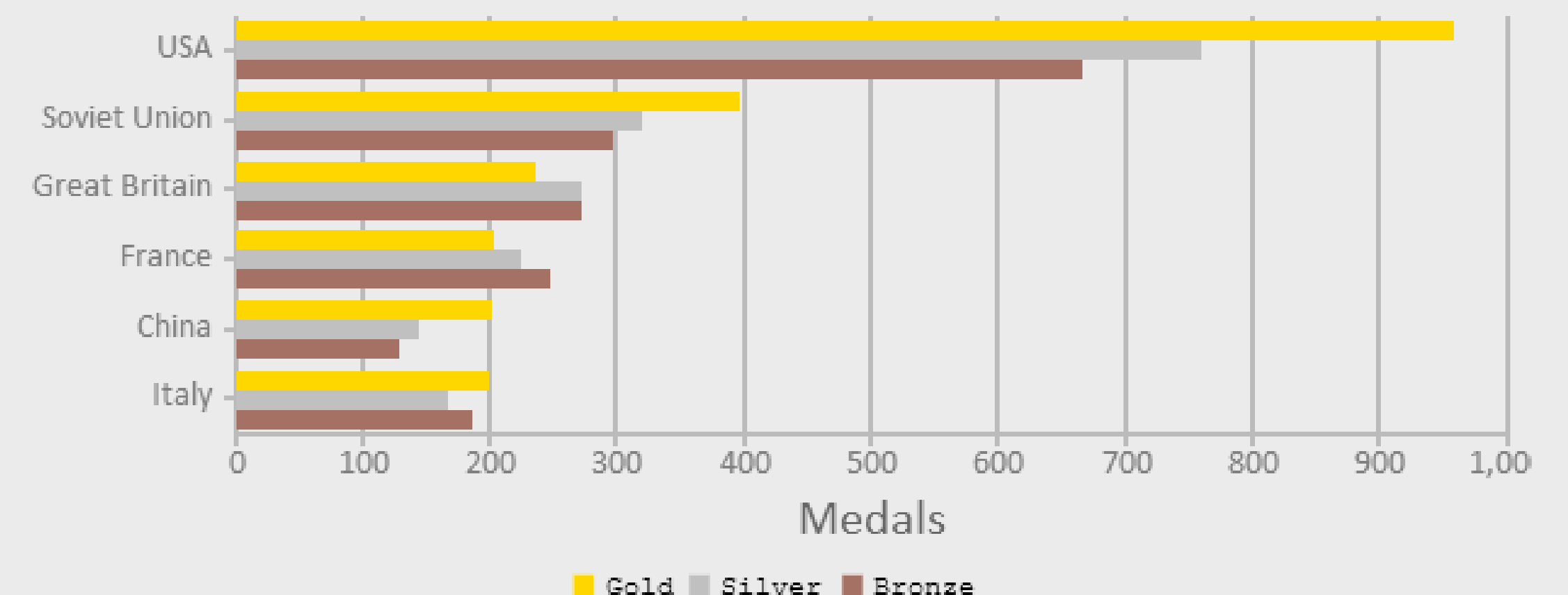


HISTOGRAM

- Is a graph display of tabulated frequencies, shown as bars
- Shows what proportion of cases fall into each of several categories
- Differs from a bar chart in that it is the area of the bar that denotes the value, not the height as in bar charts, a crucial distinction when the categories are not of uniform width
- The categories are usually specified as non-overlapping intervals of some variable. The categories (bars) must be adjacent



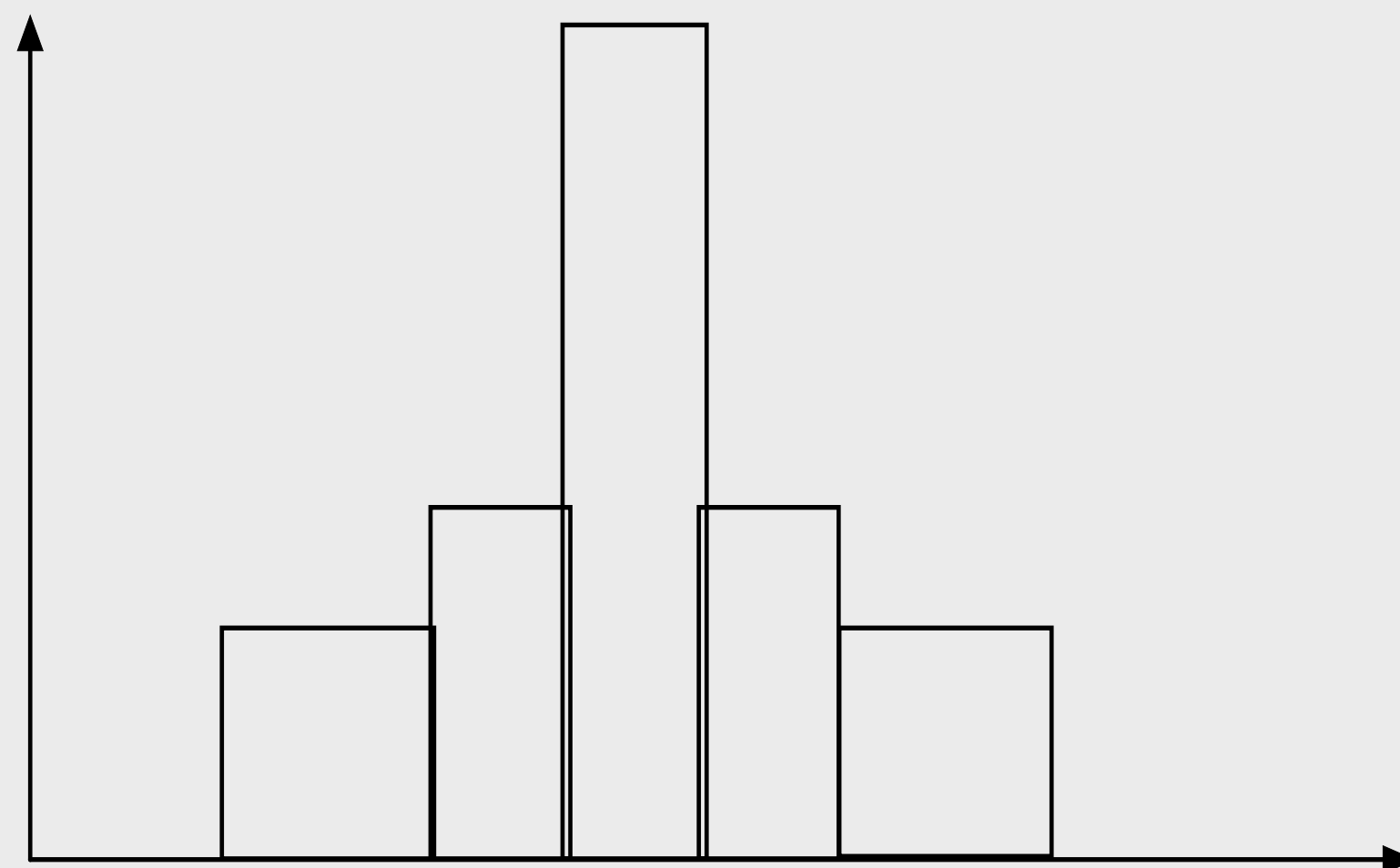
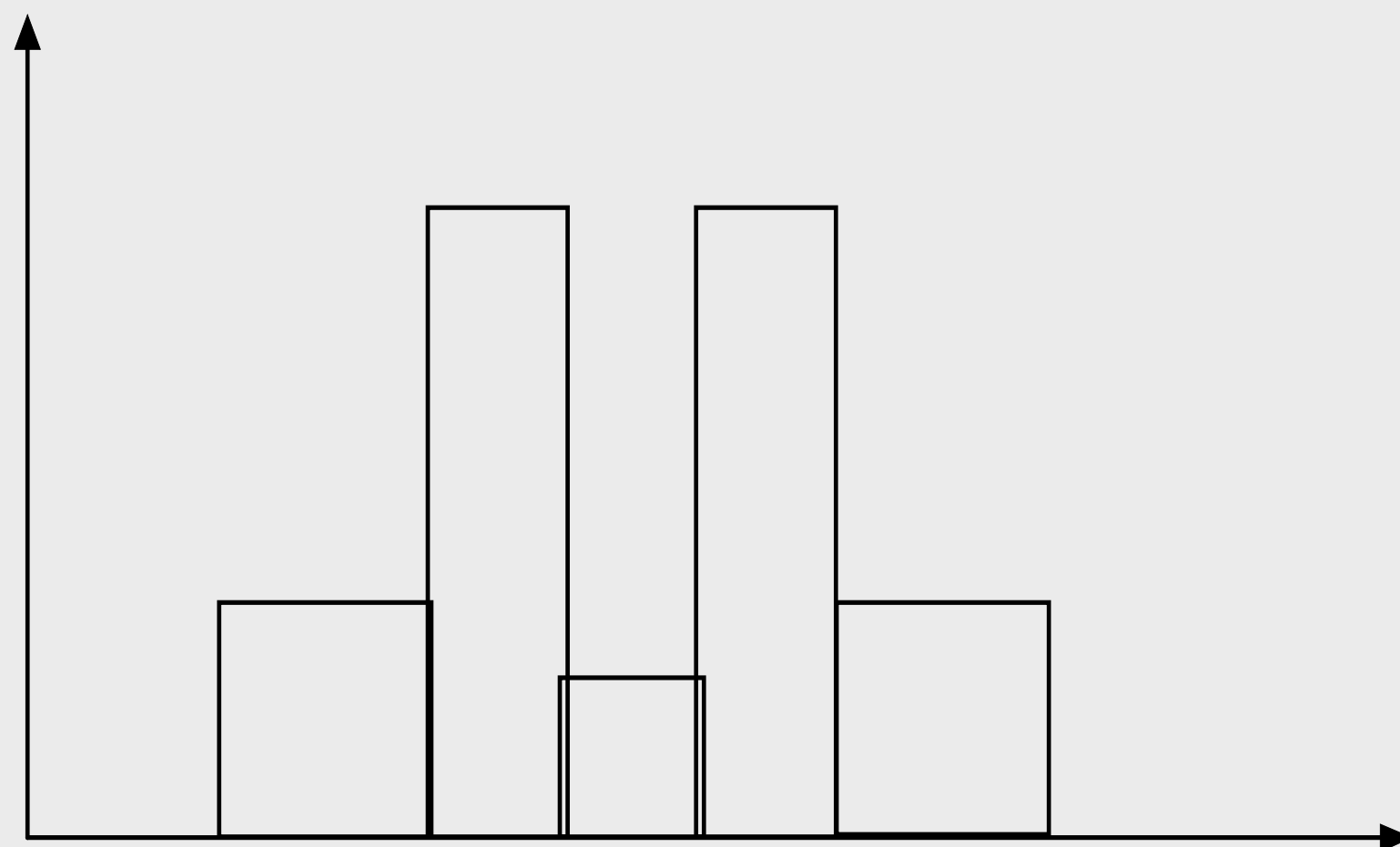
Olympic Medals of all Times (till 2012 Olympics)





HISTOGRAM (2)

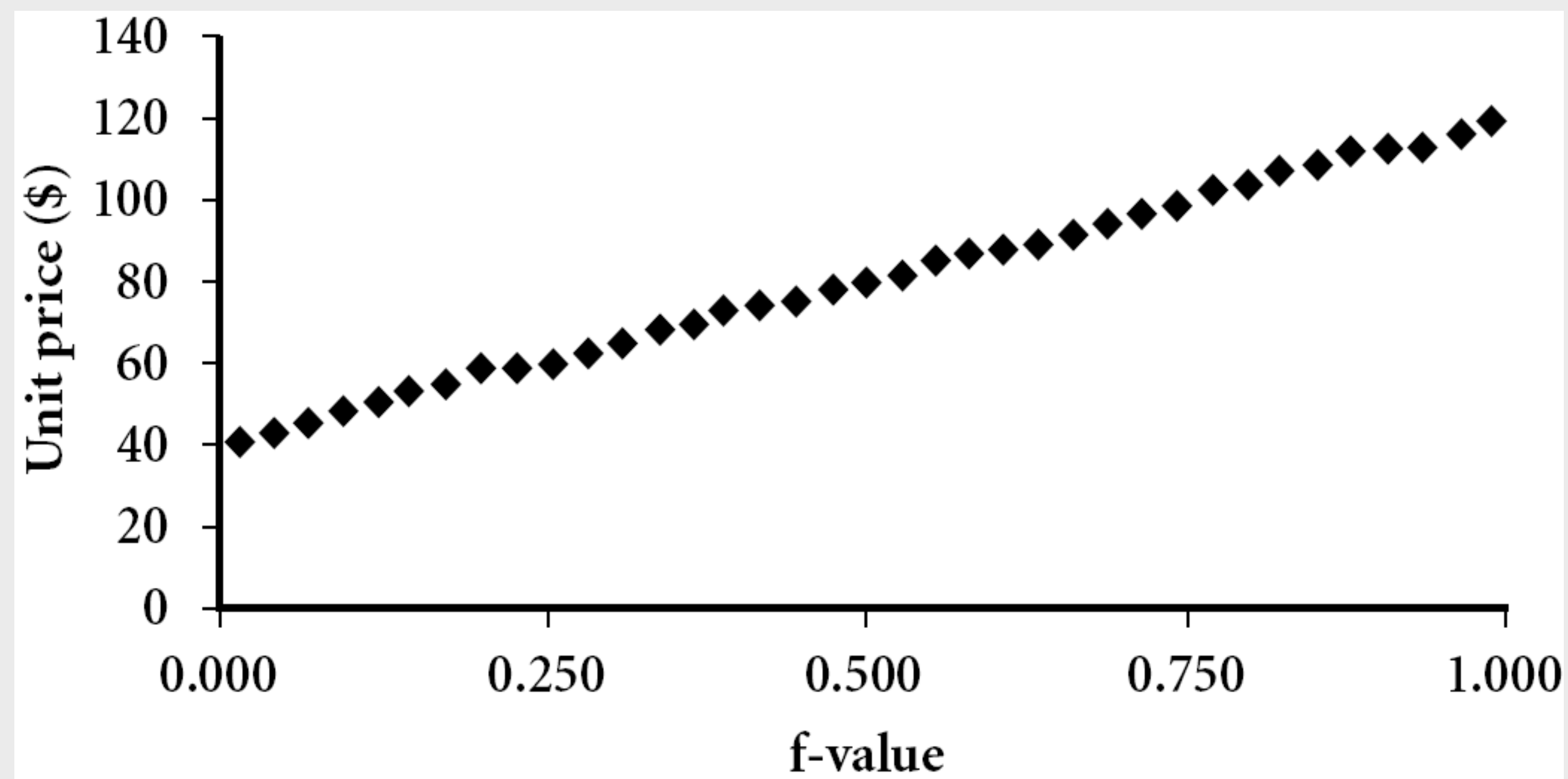
- Histograms often tell more than boxplots
- The two histograms shown in the below may have the same boxplot representation
 - The same values for: min, Q1, median, Q3, max
- But they have rather different data distributions





QUANTILE PLOT

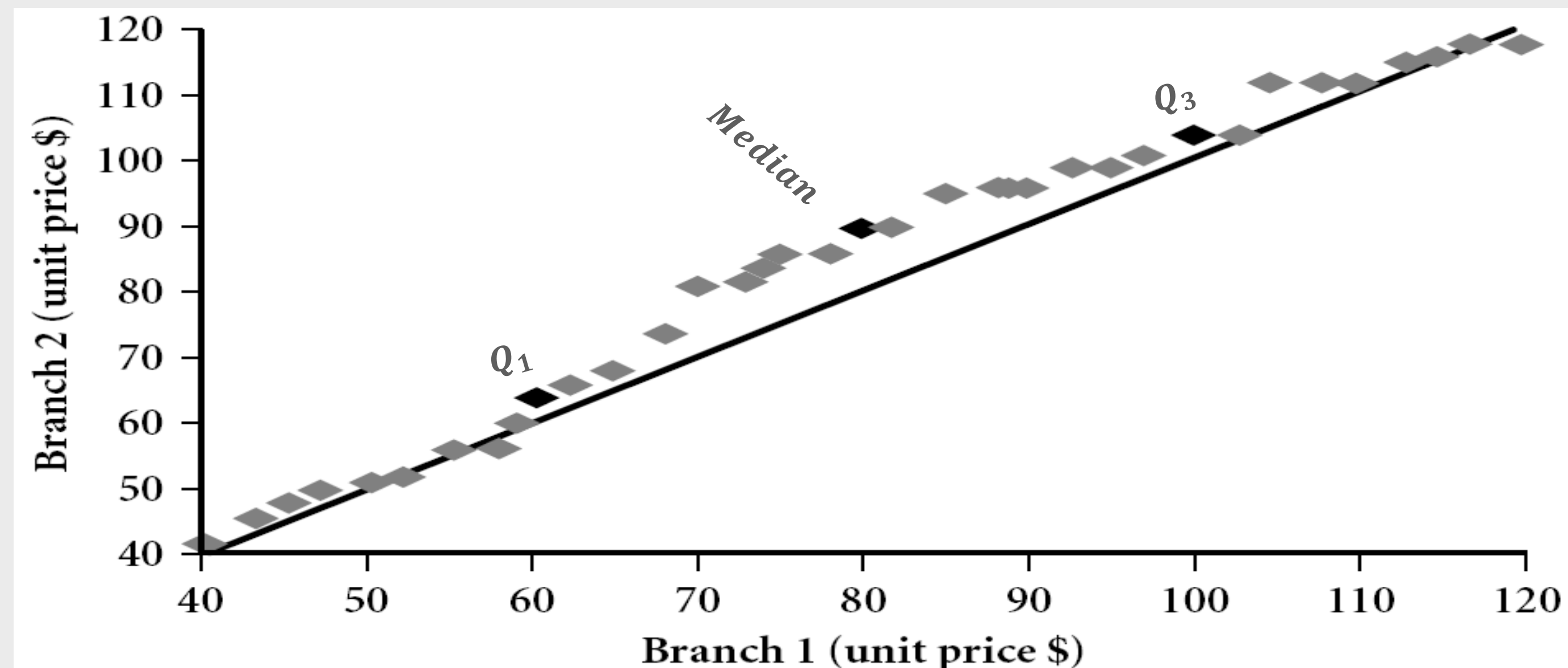
- Displays all of the data (allowing the user to assess both the overall behavior and unusual occurrences)
- Plots quantile information
 - For a data x_i , data sorted in increasing order, f_i indicates that approximately $100f_i\%$ of the data are $\leq x_i$





QUANTILE-QUANTILE (Q-Q) PLOT

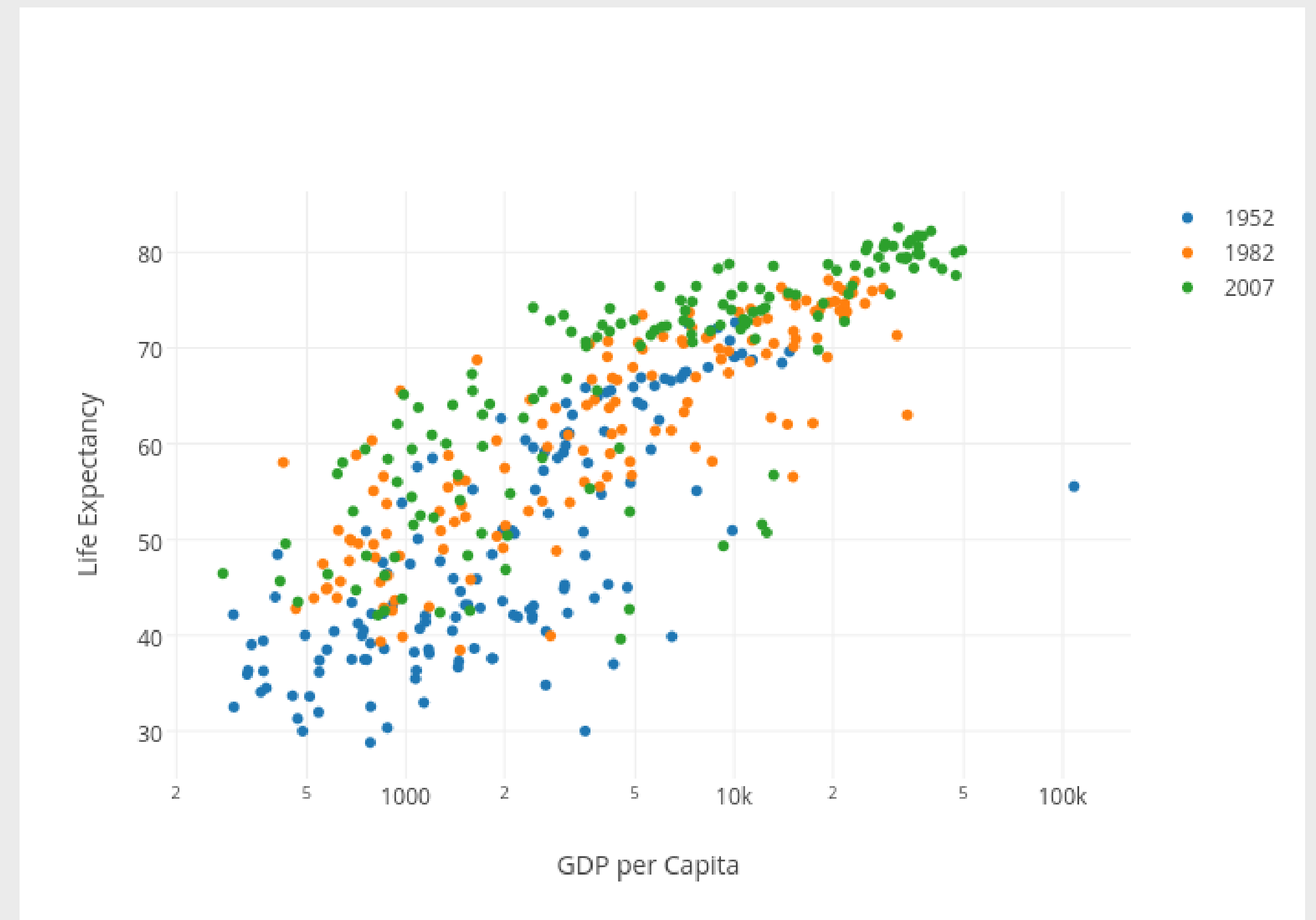
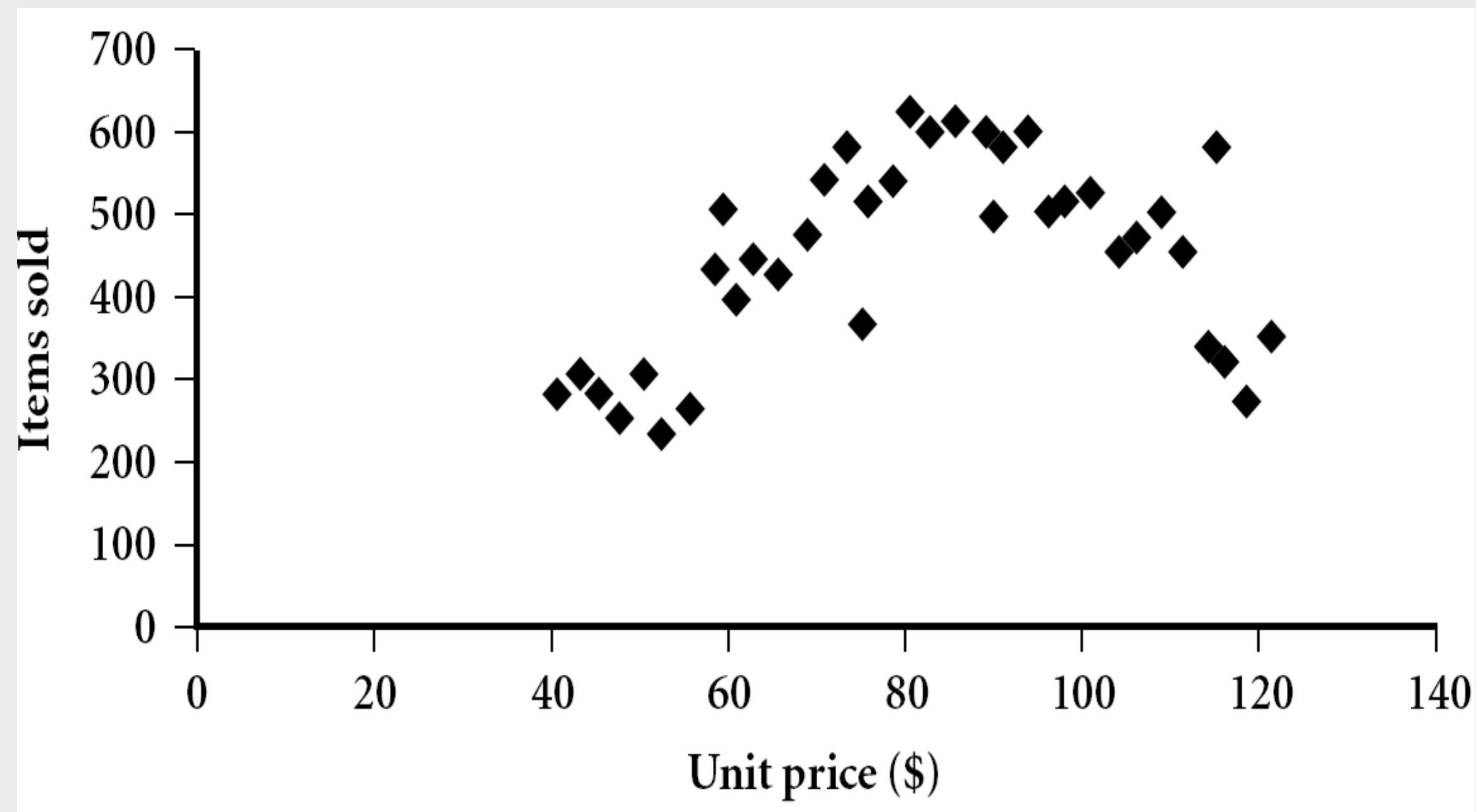
- Graphs the quantiles of one univariate distribution against the corresponding quantiles of another
- View: Is there a shift in going from one distribution to another?
- Example shows unit price of items sold at Branch 1 vs. Branch 2 for each quantile. Unit price of items sold at Branch 1 tend to be lower than those at Branch 2





SCATTER PLOT

- Provides a first look at bivariate data to see clusters of points, outliers, etc.
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane



SUMMARY

- Data is divided into two types: numeric and categorical.
- Gain insight into the data by basic statistical data description.
- Basic statistical data description: central tendency, variation, and spread

EXERCISE

1. What is the method name to gain insight into the data that involves central tendency, dispersion, and graphical displays of data?
 - a. Basic statistical data description
 - b. Data visualization
 - c. Proximity measures
 - d. Boxplot
2. Mention all kinds of attribute in a data!
3. What is the purpose of using a box plot in visualizing data?

EXERCISE (2)

3. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results:

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

Calculate the mean, median, and standard deviation of age and %fat

REFERENCES

Han, J., Kamber, M., & Pei, J. (2023). Data Mining: Concepts and Techniques (4rd ed.). San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.